

Saikumar Mandaji

EMBEDDED FIRMWARE ENGINEER · PCB DESIGNER · IOT SYSTEMS

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PROFESSIONAL SUMMARY

Embedded Firmware Engineer with 5+ years delivering production-grade firmware for medical devices, industrial IoT platforms, and smart wearables. Expertise in ARM Cortex MCUs (STM32, nRF52840, Si Labs BGM220), Bluetooth LE stack, RTOS-based multi-task architectures, and hardware-firmware co-design achieving sub-500nA sleep currents. Experience across the full product lifecycle — sensor integration, PCB design, cloud connectivity, and field deployment of 2500+ devices.

EXPERIENCE

Embedded Firmware Engineer

Apr 2024 – Present

Caropet Technologies Pvt Ltd · Hyderabad, India

- Architected firmware for **5+ medical devices** — oxygen therapy systems, therapeutic gas monitors, and PediaSense pediatric monitor — on STM32 and Si Labs BGM220, with focus on safety-critical reliability and real-time data acquisition.
- Achieved **sub-500nA sleep current** on BGM220-based BLE wearables through hardware-firmware co-optimisation — deep sleep modes, peripheral power gating, and wake-on-event architecture for multi-year coin cell operation.
- Designed custom BLE GATT profiles and beacon systems — advertising payload packing, CCCD state management, MTU negotiation, and reconnection watchdog for reliable medical fleet monitoring.
- Implemented safety-critical alarm firmware — dedicated high-priority RTOS alarm task independent of measurement tasks, hardware watchdog integration, and non-volatile alarm state persistence across power cycles.
- Designed analog front-end signal conditioning circuits for electrochemical gas sensors, SpO₂/PPG, pressure, and thermistor inputs — enabling accurate multi-sensor acquisition across medical device product lines.
- Built project-specific Python UART debug tools — real-time matplotlib plotters, binary struct parsers, BLE beacon sniffers, and automated calibration scripts — reducing firmware validation and field debugging time.

IoT & Firmware Developer

Apr 2020 – Apr 2024

Niltech Pvt Ltd · Hyderabad, India

- Built **BeetleGuard** LoRa environmental monitoring system: FreeRTOS task architecture supporting 250 sensor nodes/hub, SPI/UART peripheral drivers, AWS IoT cloud integration — **5 hubs active in production**.
- Delivered end-to-end firmware for **2500+ deployed smart card payphones** — RFID authentication, GSM calling, Raspberry Pi cloud sync, REST API, and remote fleet management dashboard.
- Developed industrial IoT data loggers on ESP32 supporting RS485/Modbus, 4–20mA, and 0–10V analog inputs with real-time MQTT cloud synchronisation.
- Prototyped ECG monitoring system on nRF52840 with physiological signal acquisition, lead-off detection, and BLE GATT streaming to a mobile app.
- Integrated TF Lite and MediaPipe ML inference on ESP32-S3 for vision-based edge applications; automated device validation with Python serial test tooling.

Embedded Systems Intern

Aug 2019 – Apr 2020

Niltech Pvt Ltd · Hyderabad, India

- Wrote register-level SPI, I²C, and UART peripheral drivers for STM32 (bypassing HAL) and AVR platforms; validated timing with logic analyser and oscilloscope.
- Contributed to multi-sensor integration, hardware bring-up, and functional test procedures across IoT, automation, and consumer electronics prototypes.

HIGHLIGHTS

5+

Years Experience

14+

Products Shipped

2500+

Devices Deployed

<500nA

Sleep Current

TECHNICAL SKILLS

MCU Platforms

STM32 nRF54L15 nRF52840 BGM220
ESP32 ATtiny85 RP2040
Raspberry Pi

Languages

Embedded C C++ Python
MicroPython

RTOS

FreeRTOS Zephyr RTOS

Wireless

BLE 5.3 LoRa/LoRaWAN Wi-Fi GSM
Thread Zigbee

Protocols

SPI I²C UART CAN RS-485
Modbus MQTT USB

PCB Design

EasyEDA Pro

Debug & Test

JTAG / SWD Logic Analyser
Oscilloscope Power Profiler
BT Protocol Analyser Python Serial

Medical Domain

Safety-Critical Firmware
Medical Alarms Analog AFE

EDUCATION

B.Tech — Electronics & Communication Engineering

Guru Nanak College of Engineering
2016 – 2019

Diploma — Electronics & Communication Engineering

Jyothishmathi Institute of Technology
2013 – 2016

KEY PROJECTS

BeetleGuard — LoRa Environmental Monitor

IOT

Scalable LoRa monitoring system; FreeRTOS task architecture for sensor polling, radio TX, and cloud sync. 250 nodes/hub · 5 hubs in field production.

STM32 · LoRa SX1278 · FreeRTOS · SPI · UART · AWS IoT

Oxygen Therapy System

MEDICAL

SpO₂ closed-loop oxygen flow control with safety-critical alarm architecture — independent RTOS alarm task, hardware watchdog, and NVM alarm state persistence.

STM32 · MAX30102 SpO₂ · Flow Control Valve · ADC · Medical Alarms · Watchdog

Therapeutic Gas Monitor (Nitric Oxide)

MEDICAL

Standalone clinical gas monitor — analog front-end for electrochemical NO sensor with safety-critical alarm thresholds and reliable concentration measurement.

STM32 · SGX-7NO · ADS1115 · MCP1525 · TIA · Medical Alarms

Patient Vitals Monitor (BLE)

MEDICAL

Ultra-low-power BLE vitals monitor on BGM220 — custom GATT profile with packed characteristic notifications; sub-500nA sleep current for multi-year battery life.

Si Labs BGM220 · BLE 5.0 · MAX30102 · NTC Thermistor · GATT · Mobile App

Industrial IoT Data Logger

INDUSTRIAL

Multi-input industrial logger supporting RS485/Modbus, 4–20mA, and 0–10V signals — live TFT display with real-time MQTT cloud synchronisation.

ESP32 · RS485 · 4–20mA · 0–10V ADC · ADS1115 · MQTT · Cloud Sync

ECG Monitoring System

MEDICAL

nRF52840-based ECG monitor — physiological signal acquisition with lead-off detection, firmware signal filtering, and real-time BLE GATT streaming to a mobile app.

nRF52840 · AD8232 · MAX30100 · DS18B20 · BLE 5.0 · ADC · Signal Conditioning

TOOLS & PLATFORMS

VS Code

STM32CubeIDE

Simplicity Studio

Segger Embedded Studio

ESP-IDF

nRF SDK

PlatformIO

Git

AWS IoT

Node-RED

MQTT Broker